Article

Factors Associated with Systolic Blood Pressure Among Middle-Aged and Older Chinese Adults

https://doi.org/na.nana/s32023-230-108-z

Recieved: 8 December 2025

Accepted: 15 December 2025

Published Online: 15 December 2025

Open access

Check for updates

Hanfei Yang [1,2], Huanqi Wu [1,2] ✉

From 2014 to 2016, a nationwide heart disease risk survey conducted by Beijing Anzhen Hospital was carried out. The life information and medical history information of 43,088 valid samples from six provinces and cities were systematically investigated. In the first stage, we identified 48 variables that had a clinical potential relationship with high systolic blood pressure, and we cleaned and coded the relevant data. In the second stage of variable selection, we employed methods such as Lasso regression, correlation analysis, and collinearity analysis, and selected 11 variables to be included in the model. We established a nested multivariate linear model based on the hypertension data of numerical variables to obtain detailed information on the contribution of specific variables. We also established a Logistic model based on binary variables to provide a reference for potential hypertension prediction. Based on the data distribution, we incorporated Box-Cox regression and power transformation for the dependent variable in the modeling process. This effectively enhanced the stability of the model and, to a certain extent, improved its interpretability. After analysis, we have confirmed that age, BMI, waist-to-hip ratio, red meat intake and fruit and vegetable consumption have a significant impact on hypertension. This helps to remind potential populations to pay more attention to various life indicators in order to control hypertension.

Introduction

Hypertension is one of the most prevalent chronic conditions in China and remains a leading risk factor for cardiovascular morbidity and mortality. As the population ages and lifestyle patterns shift, elevated blood pressure has become a critical public health challenge requiring population-level surveillance and targeted prevention strategies.

Between 2014 and 2016, Beijing Anzhen Hospital conducted a large-scale national cardiovascular risk survey that collected detailed demographic, lifestyle, dietary, anthropometric, and medical history information from 43,088 participants across six provinces. This dataset provides a unique opportunity to examine the complex relationships between lifestyle behaviors and blood pressure in a large and diverse Chinese population.

Existing large-scale epidemiological studies have primarily focused on demographic predictors and disease status, with fewer investigations addressing the combined influence of multiple lifestyle behaviors, dietary patterns, and medical history variables on systolic blood pressure (SBP) and hypertension risk. Additionally, the substantial number of candidate variables (>100) introduces challenges such as noise, multicollinearity, and heteroscedasticity. These issues hinder the development of interpretable and stable statistical models and limit the ability to quantify the relative importance of behavioral and clinical factors.

There is therefore a methodological and practical need for a systematic variable selection approach, coupled with appropriate transformations and diagnostic procedures, to construct reliable predictive and interpretive models for blood pressure outcomes.

Data Preprocessing

The original data cover multiple dimensions such as demographic characteristics, lifestyle, frequency of dietary intake, past medical history, and physical examination results. The number of variables is large and the types are complex. Therefore, before modeling, a systematic sorting, cleaning, and conversion are required to ensure that different types of variables can be uniformly, legally, and interpretably expressed in the statistical model.

During the initial data inspection stage, we first performed strict

[1] Capital University of Economics and Business, School of Statistics and Data Science. [2] University of Arizona, Department of Mathematics

Article

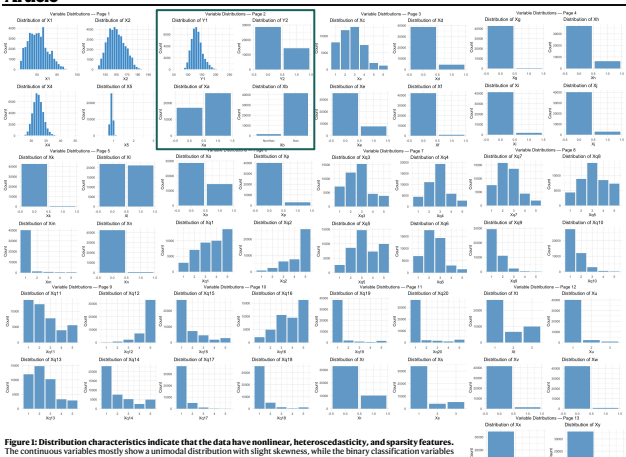


Figure 1: Distribution characteristics indicate that the data have nonlinear, heteroscedasticity, and sparsity features. The continuous variables mostly show a unimodal distribution with slight skewness, while the binary classification variables are generally extremely imbalanced. The ordered lifestyle and dietary variables exhibit a significant skewed structure, reflecting the differences in real population habits.

type conversions for all continuous variables, including age (X1), height (X2), BMI (X4), waist-to-hip ratio (X5), and systolic blood pressure (Y1). These variables entered the model directly in numeric form, which was necessary to meet the core input requirements of subsequent LASSO regression and linear regression analyses. Therefore, ensuring the correct type of these variables was the fundamental task of data preprocessing.

Subsequently, we performed standardized numerical processing on all binary categorical variables, including gender (Xa), ethnicity (Xb), smoking (Xr), and a total of 18 previous medical history variables (Xd–Xy), such as history of diabetes, heart failure, atrial fibrillation, stroke, etc. All these variables were uniformly encoded as 0 and 1, where 1 indicated "yes" or "presence of a medical history", and 0 indicated "no". This conversion method not only enhanced the model's compatibility but also made the coefficients of the subsequent linear model and Logistic model have a consistent interpretation, facilitating the inference of the impact of each disease or behavior on the risk of hypertension.

For categorical variables with strong sequential significance, such as educational attainment (Xc), sleep apnea frequency (Xm), drinking habits (Xs), and tea and coffee consumption frequency (Xt, Xu), we convert them into ordered factors (ordered factor). The categories of these variables have a natural hierarchy, but the differences do not have the meaning of continuous variables, so they cannot simply be regarded as numeric types. Processing them as ordered can retain the sequential information and avoid being misinterpreted as "quantitative differences" in linear models. In contrast, the dietary behavior variables Xq1–Xq22 have a relatively clear hierarchical meaning (the intake frequency changes from low to high), and they are often regarded as approximate continuous lifestyle indicators in actual epidemiological studies. In this study, they are converted into numeric variables ranging from 1 to 5, with the larger the number representing more frequent intake, enabling these variables to effectively participate in screening in LASSO.

To ensure the correctness of linear models, this study conducted strict handling of missing values for all variables during the data preprocessing stage. Due to the large volume of data and the relatively low proportion of missing values, and because Box-Cox requires a complete continuous dependent variable, this study employed complete-case analysis (na.omit) to eliminate records containing missing values, resulting in a final model sample that is complete and meets the mathematical assumptions.

In the distribution examination of the continuous dependent variable systolic blood pressure (Y1), we observed that it had right skewness, variance that increased with the value, and other issues, which violated the assumption of independent and identically distributed error terms in the linear model. Therefore, in this study, we applied the Box–Cox transformation to Y1 to improve its distribution characteristics. After obtaining the optimal λ through maximum likelihood estimation, we recorded the transformed result as Y1_bc and used it to replace the original Y1 in subsequent model analyses. This processing significantly improved the approximate normality and variance stability of the dependent variable, thereby enhancing the model's fitting performance and explanatory power.

$$Y_{1,bc} = \frac{Y_1^{\lambda^*} - 1}{\lambda^*}$$

After completing the variable type conversion, encoding, missing value handling, and Box-Cox transformation, we obtained final analysis data that are structurally clear, of uniform type, and can

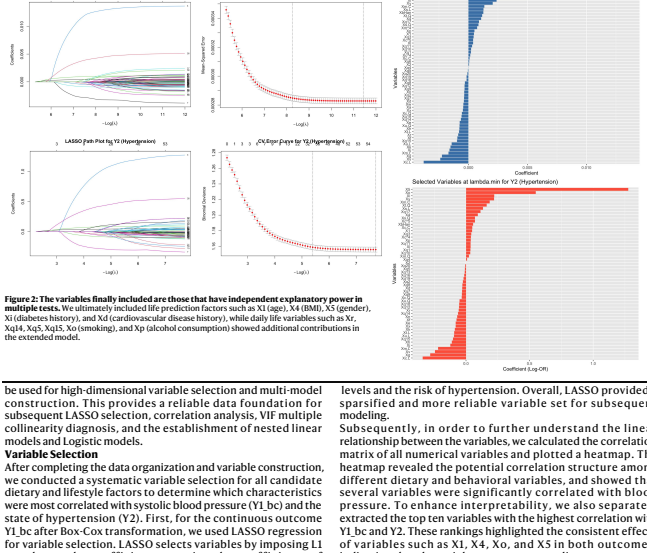


Figure 2: The variables finally included are those that have independent explanatory power in multiple tests. We ultimately included life prediction factors such as X1 (age), X4 (BMI), X5 (gender), Xi (diabetes history), and Xd (cardiovascular disease history), while daily life variables such as Xr, Xq14, Xq5, Xq15, Xo (smoking), and Xp (alcohol consumption) showed additional contributions in the extended model.

be used for high-dimensional variable selection and multi-model construction. This provides a reliable data foundation for subsequent LASSO selection, correlation analysis, VIF multiple collinearity diagnosis, and the establishment of nested linear models and Logistic models.

Variable Selection

After completing the data organization and variable construction, we conducted a systematic variable selection for all candidate dietary and lifestyle factors to determine which characteristics were most correlated with systolic blood pressure (Y1_bc) and the state of hypertension (Y2). First, for the continuous outcome Y1_bc after Box-Cox transformation, we used LASSO regression for variable selection. LASSO selects variables by imposing L1 penalty on the coefficients, causing the coefficients of unimportant variables to shrink to zero, thereby achieving automated variable selection in high-dimensional scenarios. After determining the optimal λ (i.e., λ_min) using five-fold cross-validation, the model showed stability in the path diagram and error curve, and finally selected a set of variables with the highest explanatory power. The bar chart of LASSO results visually presents the coefficient sizes of the selected variables, indicating that variables such as X5, Xo, and Xi contribute significantly to predicting systolic blood pressure.

$$\hat{\beta} = \arg\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} (y_i - x_i^{\top}\beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

$$\hat{\beta} = \arg\min_{\beta} \left\{ -\frac{1}{n} \sum_{i=1}^{n} [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Similarly, we used the logistic LASSO model to analyze the binary outcome Y2 (whether one has hypertension). Since the coefficients of this model can be interpreted as log-odds increments, it can be used to identify which lifestyle factors may increase the risk of the disease. The cross-validation results also showed that the model was stable. The variable selection graph at lambda.min indicated the most critical influencing factors, and some variables repeatedly appeared in both the continuous and binary models, suggesting their joint influence on blood pressure levels and the risk of hypertension. Overall, LASSO provided a sparsified and more reliable variable set for subsequent modeling.

Subsequently, in order to further understand the linear relationship between the variables, we calculated the correlation matrix of all numerical variables and plotted a heatmap. The heatmap revealed the potential correlation structure among different dietary and behavioral variables, and showed that several variables were significantly correlated with blood pressure. To enhance interpretability, we also separately extracted the top ten variables with the highest correlation with Y1_bc and Y2. These rankings highlighted the consistent effects of variables such as X1, X4, Xo, and X5 in both outcomes, indicating that they might represent some dietary patterns or lifestyle characteristics closely related to blood pressure.

$$r_{XY} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

During the variable selection process, we also had to ensure that the model did not have severe multicollinearity. Therefore, the VIF index was calculated for both the linear model (used for Y1_bc) and the logistic model (used for Y2). The results showed that the GVIF^(1/(2*Df)) of all variables was far below the commonly used thresholds (such as 5 or 10), indicating that there were no collinearity issues among the variables that might affect the stability of the estimates. Since there were no variables with VIF exceeding the threshold, there was no need to eliminate or further process them. This result also indirectly confirmed that the variable selection results obtained by LASSO were robust and not affected by high collinearity.

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}$$

In summary, the variable selection part established a clear feature framework through LASSO, correlation analysis, and collinearity diagnosis, enabling us to identify the dietary and lifestyle variables that have the greatest impact on blood

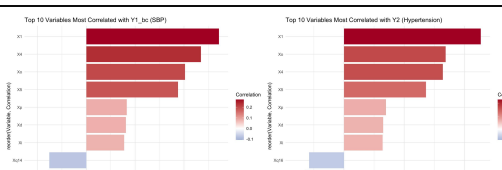


Figure 3: Top 10 Correlation factor

pressure and the risk of hypertension. These analyses laid a reliable and scientific variable foundation for the subsequent multiple regression model, and ensured that the model construction process was statistically reasonable and highly interpretable.

Modeling and Preliminary Analysis

After completing the variable selection, based on the results of the previous LASSO, correlation analysis, and VIF tests, we constructed three regression models to systematically evaluate the effects of individual characteristics, medical history, and lifestyle factors on systolic blood pressure levels (SBP) and the risk of hypertension. Considering that age and weight indicators may exhibit nonlinear effects, this study introduced the quadratic term of age X_1^2 and the third-order natural spline of BMI in all models. At the same time, considering that numerical variables have inconsistent units, in order to balance the influence of each parameter, we adopted the Zscore compression method (marked as Xn_z in the code, and still referred to as Xn in the following text) for them.

$$Z_i = \frac{X_i - \mu_X}{\sigma_X}$$

ns(X4, 3) is used to more fully capture the potential physiological change trends. The model gradually expands based on different research purposes, making the analysis structure both rigorous and interpretable.

model 1: basic linear model

$$\begin{aligned} Y1 = {} & \beta_0 + \beta_1 X1 + \beta_2 X1^2 + \beta_3\, ns(X4)_1 + \beta_4\, ns(X4)_2 + \beta_5\, ns(X4)_3 \\ & + \beta_6 X5 + \beta_7 Xo + \beta_8 Xp + \beta_9 Xi + \beta_{10} Xd + \varepsilon. \end{aligned}$$

Firstly, Model 1 (the basic linear model of SBP shows above) mainly includes core variables such as age, BMI, gender, alcohol consumption, history of diabetes, and history of cardiovascular diseases, to depict the influence of basic individual physiological and medical history factors on SBP. The model results show that all main variables are at a significant level (p < 0.01), among which age shows a positive but gradually weakening non-linear effect, and each spline component of BMI is significant, indicating that its relationship with SBP is clearly non-linear; male gender, fat intake (Xo), and history of diabetes (Xi) all have significant positive effects on SBP. The R^2 of this model is 0.512, and adjR^2 is 0.514, it has a significant improvement in explanatory power compared to the results without Z-score compression..

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

Based on this, Model 2 (SBP Extended Linear Model) further incorporated the key lifestyle variables that repeatedly appeared in LASSO (Xq14, Xq5, Xq15) and smoking status (Xr) to test the additional contribution of dietary structure and behavioral patterns to blood pressure. The results showed that the three newly added dietary/behavioral variables were all highly significant. Among them, Xq14, Xq5, and Xq15 all had negative coefficients, suggesting that certain dietary or habit factors have a protective effect on blood pressure. With the addition of these variables, the model's explanatory power increased to R^2 = 0.70, which was significantly improved compared to the basic model, indicating that lifestyle plays an important role in blood pressure regulation.

mode 2: extended linear model

$$\begin{aligned} Y1 = {} & \beta_0 + \beta_1 X1 + \beta_2 X1^2 + \beta_3\, ns(X4)_1 + \beta_4\, ns(X4)_2 + \beta_5\, ns(X4)_3 \\ & + \beta_6 X5 + \beta_7 Xo + \beta_8 Xp + \beta_9 Xi + \beta_{10} Xd \\ & + \beta_{11} Xr + \beta_{12} Xq14 + \beta_{13} Xq5 + \beta_{14} Xq15 + \varepsilon, \end{aligned}$$

We applied a natural spline transformation to BMI (X4) in order to allow for a more flexible and non-linear relationship between it and systolic blood pressure:

$$\text{ns}(X4, 3) = \{B_1(X4),\ B_2(X4),\ B_3(X4)\}$$

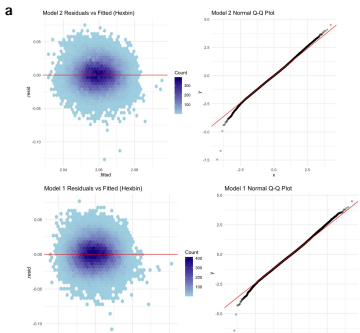
Here, ns (X4) represents the spline basis function, which satisfies the natural constraint that the second derivatives at both ends are zero. In actual biomedical data, the relationship between BMI and SBP is usually not linear: a low BMI has a relatively weak association with SBP, while the influence is stronger in the overweight range. This makes it difficult for simple linear or quadratic terms to accurately capture this structure. Among the coefficients corresponding to the three spline basis functions, the t-statistics are all highly significant (in Model 1, t = 33.25, 12.09, 7.27, p < 0.001). Based on the possible non-linear structure, we retained this adjustment after a series of comparisons.

model 3: logistic regression linear model

$$\begin{aligned} \log\left(\frac{P(Y2=1)}{1-P(Y2=1)}\right) = {} & \beta_0 + \beta_1 X1 + \beta_2 X1^2 + \beta_3\, ns(X4)_1 + \beta_4\, ns(X4)_2 + \beta_5\, ns(X4)_3 \\ & + \beta_6 X5 + \beta_7 Xo + \beta_8 Xi + \beta_9 Xp + \beta_{10} Xd + \beta_{11} Xq16 + \beta_{12} Xq5 + \beta_{13} Xq15. \end{aligned}$$

Finally, Model 3 (the Hypertension Logistic Model) uses whether having hypertension (Y2) as the dependent variable and employs the Logit link function for modeling. In terms of variable composition, it continues the core terms of the previous two linear models and additionally incorporates LASSO significant lifestyle factors (Xq16, Xq5, Xq15). The results show that age, BMI splines, gender, and fat intake all significantly increase the risk of hypertension; while some dietary/behavioral variables (Xq16, Xq5, Xq15) have a significant negative effect, consistent with the SBP model, possibly reflecting a healthier lifestyle. This model converges rapidly (4 iterations), with an AIC of 50,290, and has good fit.

Overall, the three models jointly revealed the complexity of blood pressure regulation: age and BMI are the most stable physiological influencing factors, while gender and history of



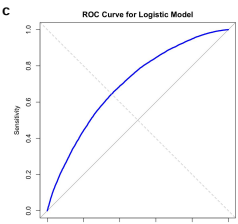


Figure 4: a. Model Diagnostics The residual plots and qq plots of the two models exhibited normal residual patterns. **b. Significant factors** The table below presents the significant factors in the three models. Due to the large sample size, we adopted the criterion of $p < 0.001$ for judgment. **c.Prediction model ROC** The ROC curve shows no abnormalities and the predictive ability is moderate.

b

	Model 1 Estimate	Std.Error	p-value	Model 2 Estimate	Std.Error	p-value	Model 3 Estimate	Std.Error	p-value
X1 Age	9.041e-04	9.47e-05	<2e-16	8.595e-04	9.38e-05	<2e-16	0.100144	0.012756	2.85e-15
X5 Sex (1 for Male)	1.814e-02	1.13e-03	<2e-16	1.489e-02	1.13e-03	<2e-16	1.373916	0.147848	<2e-16
X6 Medicine	3.857e-03	2.19e-04	<2e-16	4.400e-03	2.09e-04	<2e-16	0.461087	0.026404	<2e-16
X1 Diabetes	2.223e-03	4.36e-04	9.13e-08	2.125e-03	4.12e-04	2.51e-07	0.184764	0.050928	2.86e-04
Xq14 Milk	—	—	—	-9.017e-04	6.65e-05	<2e-16	—	—	—
Xq5 Red Meat	—	—	—	-4.642e-04	6.92e-05	1.99e-11	-0.052216	0.009218	1.47e-08
Xq15 Yogurt	—	—	—	-1.386e-03	7.67e-05	<2e-16	-0.171154	0.009977	<2e-16
Xq16 Fruit	—	—	—	—	—	—	-0.059092	0.009361	2.74e-10

metabolic diseases also have consistent and significant effects; and some dietary and lifestyle variables show robust contributions in the SBP and hypertension risk models, suggesting their potential independent regulatory roles. This stage of modeling has laid a solid statistical foundation for us to give useful subsequent medical explanations, intervention recommendations, and risk stratification analyses.

Model Diagnostics

After establishing the three types of regression models, we conducted a systematic diagnostic analysis of the model performance and the satisfaction of the assumptions. For Model 1 and Model 2 (the linear regression model of SBP), the residual-fitted plot (Residuals vs Fitted) was presented in the form of hexbin density. The results showed that the residuals were uniformly distributed around 0, and no obvious funnel-shaped pattern was observed as the fitted value increased, indicating that the homoscedasticity assumption was basically satisfied. This further confirmed the corrective effect of the BoxCox regression on the dependent variable. Additionally, in the normal Q-Q plots of the two groups of models, most of the standardized residuals were in good agreement with the theoretical normal distribution, with only a slight deviation in the tail, which is a common phenomenon in large samples and does not affect the stability of the model.

For Model 3 (Hypertension Logistic Regression), we evaluated the classification performance through the ROC curve. The result showed that the AUC = 0.696, indicating that the model has a moderate level of discrimination ability in distinguishing hypertensive individuals from non-hypertensive individuals. Further, based on the default threshold of 0.5, a confusion matrix was constructed. The overall accuracy of the model was 0.686, with a relatively high specificity (0.897) and a relatively low sensitivity (0.65). This indicates that the model performed better in identifying non-hypertensive individuals. This phenomenon is consistent with the sample structure of uneven hypertension prevalence (approximately 33%), reflecting that the model is affected by class imbalance under the current threshold. The visualization of the confusion matrix further emphasizes this feature: most of the predicted "0" came from the true negative category, while the detection rate of the positive class was relatively limited, suggesting that if the research purpose focuses on improving the positive detection rate, subsequent adjustments to the threshold or the introduction of cost-sensitive learning can be considered.

In the extended analysis of the linear model, we further utilized LASSO to select all possible pairwise interaction terms. The results showed that several interaction terms had non-zero coefficients, especially terms such as X5:Xi, Xi:Xd, X0:Xd, etc., with relatively large absolute coefficients. This suggests that there is a non-linear interaction effect among the obesity indicators (waist-to-hip ratio), history of stroke, and history of cardiovascular metabolic diseases, and the influence on SBP is not solely explained by the main effect. Additionally, multiple dietary variables also exhibited stable interaction effects with core clinical variables (such as BMI, chronic disease medication, smoking), indicating that dietary behaviors may jointly affect

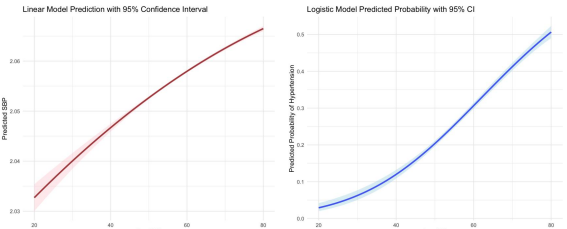


Figure 5: Confidence Interval for two types of model

blood pressure levels by modifying past diseases or lifestyle factors. Through this interaction term screening, it can provide important basis for the subsequent nonlinear extension of the model and the exploration of complex health behavior mechanisms.

Overall, the key variables provide us with stable and reliable explanations for SBP; the Logistic model has moderate discriminative ability, but its classification performance still has room for further improvement in the case of imbalanced classes; and the introduction of interaction terms reveals a more complex health behavior-disease mechanism, providing an expansion direction for subsequent work.

ANOVA nested model comparison

After completing the baseline linear model (Model 1) and the extended linear model (Model 2) for SBP, we further used the ANOVA of the nested models to test whether the new variables (Xr, Xq14, Xq5, Xq15) added in Model 2 could significantly enhance the explanatory power of the model for systolic blood pressure. Since Model 1 is a strict sub-model of Model 2 (that is, Model 2 only adds four variables related to diet and smoking on the basis of Model 1), the goodness of fit of the two models can be compared through the F-test of ANOVA.

$$F = \frac{(RSS_1 - RSS_2)/(df_1 - df_2)}{RSS_2/df_2}$$

The comparison of nested models is based on the F-test. Our hypothesis is :

H0: β of the new variable equals 0
H1: β of the new variable is not equal to 0

The analysis results show that the F value given by ANOVA is 241.64, and the corresponding p-value is less than 2.2×10^{-16}, which is much lower than the commonly used significance level, indicating that Model 2 is statistically significantly superior to Model1.

This result indicates that lifestyle-related variables such as dietary frequency and smoking status (Xr, Xq14, Xq5, Xq15) not only showed potential contributions in the previous LASSO and correlation analysis, but also significantly improved the explanatory power of the model after being incorporated into the linear model. In other words, compared to the basic model that only uses demographic characteristics and past medical history, including these lifestyle variables can more fully capture the variation structure of SBP, making the model more complete and accurate in explaining changes in systolic blood pressure. Therefore, from both a statistical significance and model performance perspective, the extended model (Model 2) is significantly superior to the basic model, which also supports the necessity of continuing to include lifestyle factors in subsequent multivariate analyses.

Prediction and Confidence Intervals

After completing variable selection and model construction, we further utilized the research data to evaluate the predictive ability of the model.

In the linear model section, we constructed a new prediction dataset based on Model 1, only changing the variable X1 (age), while keeping the other variables at the typical levels of the sample (median or 0), thereby generating a SBP prediction curve that varies with age. Using predict(..., interval = "confidence"), we obtained the 95% confidence intervals (CI) for each age point and plotted them as a linear prediction graph. This graph shows a smooth increase trend of systolic blood pressure with age, with a slightly curved curve, reflecting the non-linear contributions of the quadratic term and the BMI spline function in the model. The narrow confidence intervals indicate that the model estimation is stable and has a small variance, representing a good and medically logical prediction result.

For the risk of hypertension (Y2), we generated a prediction probability curve through the logistic model. Using type="response", we converted the logit prediction into probability values, and calculated the confidence intervals on the probability scale using the standard error of linear prediction. The graph shows the predicted probability of hypertension within the age range from 20 to 80 years and its 95% CI. The curve presents a typical cardiovascular risk pattern that rises rapidly with age, especially being steeper in the 50–60 age range, indicating that the model successfully captures the non-linear influence of age on hypertension. The confidence bands are narrow, indicating high probability estimation accuracy.

From the prediction results, whether it is continuous blood pressure or binary hypertension risk, the model shows a reasonable, smooth, and highly interpretable trend, with stable confidence intervals and no abnormal shapes, indicating a good model fit and reliable predictive ability. These prediction graphs not only demonstrate the model performance but also provide intuitive tools for subsequent health risk assessment. For example, we can estimate the risk levels for different age groups and support the design of interventions at the clinical or public health level.

Conclusion

This study is based on the multi-provincial cardiovascular risk survey data of 43,088 Chinese middle-aged and elderly residents. Through variable selection, Box-Cox transformation, multi-model construction, and predictive analysis, it systematically evaluated the effects of demographic, physical indicators, lifestyle, and previous disease history on systolic blood pressure

(SBP) and hypertension. The data cover multi-dimensional information such as lifestyle, dietary frequency, physical measurements, and disease history, providing a reliable clinical and epidemiological basis for analysis.

After data cleaning, we used Box-Cox transformation on SBP to meet the assumptions of the linear model, and employed LASSO to complete the selection of high-dimensional variables. The results were consistently identified as core influencing factors in both the continuous SBP model and the binary hypertension model: age (X1), BMI (X4), waist-to-hip ratio (X5), gender, smoking (Xo), alcohol consumption (Xp), diabetes history (Xi), and cardiovascular disease history (Xd) all had significant effects; some dietary behaviors (such as Xq14, Xq5, Xq15) showed protective effects.

We then constructed three types of regression models:

Model 1 (basic linear model) added the quadratic term of age and the spline of BMI (ns(X4,3)) to capture the physiological non-linear trend. The results showed that age was continuously positively correlated with SBP, the relationship between BMI and SBP was non-linear, and previous disease history significantly increased SBP ($R^2 = 0.512$).

Model 2 (extended model) included the lifestyle variables selected by LASSO, significantly improving the explanatory power to $R^2 = 0.70$. ANOVA showed that the model improvement was highly significant ($F = 241.64$, $p < 2.2\times10^{-16}$), indicating that dietary and smoking lifestyle factors had additional and independent explanatory power for SBP.

Model 3 (Logistic hypertension model) showed that the probability of hypertension was approximately 3% at the age of 20 and increased to over 50% at 80 years old. Age, BMI spline, smoking, diabetes history, etc. all significantly increased the risk of disease, while dietary behaviors showed protective effects. The model AUC = 0.696, with moderate predictive performance.

Based on the results of this study, Model 2 (the extended linear model) is the model with the best overall performance. Firstly, from the perspective of statistical performance, the explanatory power of Model 2 is significantly higher than that of the base model (R^2 increased from 0.512 to 0.70), and it is strongly supported by nested ANOVA ($F = 241.64$, $< 2.2\times10^{-16}$). This significant improvement not only reflects the improvement in model fit, but also indicates that variables such as lifestyle and dietary frequency have an independent and non-negligible contribution in explaining the fluctuations of SBP.

Based on all the results, this study forms the following core medical conclusions:

1. Age and BMI are the most stable and strongest determinants of SBP/hypertension, with non-linear effects.
2. Smoking, alcohol consumption, diabetes history, and cardiovascular disease history significantly increase SBP and hypertension risk, being key intervention targets.
3. Dietary behavior factors have significant independent contributions, indicating that improving dietary structure (such as reducing high-fat intake, increasing specific healthy dietary frequency) can play a positive role in controlling blood pressure.
4. The multi-model consistency is strong, and both continuous and binary models repeatedly identified the same core variables, enhancing the credibility of the results.

These conclusions not only have statistical significance but also conform to the clinical medical theory on the comprehensive regulation of metabolism-inflammation-vascular system, providing valuable evidence for hypertension risk screening and individualized intervention in the field of public health.

Limitation

There are still several limitations that need to be noted. Firstly, the data is derived from cross-sectional surveys, so it is impossible to directly infer causal relationships.

Many of the observed significant effects (such as the non-linear relationship between BMI and SBP) may be influenced by unmeasured confounding factors (such as genetic factors, medication adherence, and long-term dietary pattern stability). Future studies should use more systematic methods for handling missing data (such as multiple imputation).

In terms of statistical methods, although LASSO helps to screen high-dimensional features, its results may still be affected by the variable coding method and the λ selection method; moreover, LASSO cannot directly handle the substitution relationships between strongly correlated variables, so some variables may be excluded due to high correlation, thereby underestimating their true clinical significance. The ROC of the model (0.696) indicates moderate predictive ability, suggesting that there is still room for improvement in capturing complex lifestyle risk patterns. Additionally, due to the seasonality of dietary and behavioral variables, self-report bias, and ambiguity (such as intake frequency not reflecting actual intake), the model's estimation of lifestyle may be biased.

Finally, the study did not include follow-up data, family genetic history, specific medication use, renal function indicators, and other key medical variables, thus the explanation of the mechanism of hypertension remains incomplete.

Future Work

Based on the findings and limitations of this study, further extension work can be carried out at multiple levels. Firstly, in terms of statistical methods, more nonlinear and interaction effect-enhanced models such as Generalized Additive Models (GAM), Random Forest, Gradient Boosting, etc. can be used to improve the predictive ability and more comprehensively capture the complex interaction patterns of lifestyle factors. At the same time, methods such as LASSO-interaction and hierarchical Bayesian models can be used to systematically identify the interaction structures between variables and verify whether the selected interaction terms have deeper biological significance.

At the data level, if follow-up data can be obtained in the future, time series models or survival models for hypertension occurrence, such as Cox regression or Machine-learning-Survival models, can be constructed to achieve causal inference and risk trajectory analysis. In addition, more clinical variables such as blood lipids, renal function, inflammatory indicators (CRP), exercise intensity, sleep quality, etc. should be included to increase the explanatory power of the model. For dietary variables, continuous quantitative methods (such as precise nutrient intake) can be used to reduce self-report bias and improve estimation reliability.

From the perspective of medical application, future research should attempt to develop a hypertension risk stratification system based on model prediction results to guide early warning and individualized intervention strategies. For example, a focused screening plan can be established in the rapid risk increase interval of 50-60 years old, and lifestyle intervention guidance tools can be developed using dietary behavior variables. In conclusion, future research should continue to optimize statistical modeling techniques and integrate more comprehensive medical indicators and follow-up information to make hypertension risk prediction more accurate, providing a more scientific basis for individualized health management and public health decision-making for Chinese residents.

Code availability

The core functions of the code are presented in Appendix A at the end. We are also actively considering uploading the results to GitHub.

Data availability

Due to the patient's personal information, we are unable to provide the complete dataset. However, we are more than willing to offer a dataset section to facilitate the presentation of the initial data structure, which is demonstrated in Appendix B.

Article

Appendix A

Note: Here we present the core codes that we consider to be crucial for the encoding, modeling and diagnosis stages. Some image codes and transformation codes have been excluded as they are not related to the core functions or were not included in the report. Any further questions regarding the codes will be addressed.

```
library(dplyr)
library(tidyr)
library(glmnet)
library(ggplot2)
library(ggcorrplot)
library(gridExtra)
library(patchwork)
library(car)
library(pROC)
library(splines)
library(MASS)


hypertension <-
read.csv("/Users/huanqiwu/Desktop/467FinalData.cs
v")

hypertension$Xa <- as.numeric(hypertension$Xa)

hypertension$Xb <- factor(hypertension$Xb,
levels=c(0,1), labels=c("NonHan","Han"))

binary_vars <-
c("Xd","Xe","Xf","Xg","Xh","Xi","Xj","Xk","Xl",

"Xn","Xo","Xp","Xr","Xv","Xw","Xx","Xy")
hypertension[binary_vars] <-
lapply(hypertension[binary_vars], as.numeric)

diet_vars <- paste0("Xq",1:20)
hypertension[diet_vars] <-
lapply(hypertension[diet_vars], as.numeric)

hypertension$Xc <- ordered(hypertension$Xc,
levels=1:6)
hypertension$Xm <- ordered(hypertension$Xm,
levels=1:5)
hypertension$Xs <- ordered(hypertension$Xs,
levels=1:3)
hypertension$Xt <- ordered(hypertension$Xt,
levels=1:3)
hypertension$Xu <- ordered(hypertension$Xu,
levels=1:3)


num_vars <- c("X1","X2","X4","X5","Y1")
hypertension[num_vars] <-
lapply(hypertension[num_vars], as.numeric)

hypertension$Y2 <- as.numeric(hypertension$Y2)

tmp <- na.omit(hypertension)
bc <- boxcox(Y1 ~ 1, data=tmp)
lambda <- bc$x[which.max(bc$y)]

hypertension$Y1_bc <- (hypertension$Y1^lambda - 1)
/ lambda
hypertension_clean <- na.omit(hypertension)
hypertension_clean$Y1 <- hypertension_clean$Y1_bc

y_y1 <- hypertension_clean$Y1_bc
predictors_y1 <-
setdiff(names(hypertension_clean),
c("Y1","Y1_bc","Y2","X3"))
x_y1 <- model.matrix(~ .,
data=hypertension_clean[predictors_y1])[,-1]

set.seed(123)
lasso_y1 <- cv.glmnet(x_y1, y_y1, alpha=1,
family="gaussian")

coef_df_y1 <- data.frame(
  variable = rownames(coef(lasso_y1,
s="lambda.min")),
  coefficient = as.numeric(coef(lasso_y1,
s="lambda.min"))
) %>% filter(variable!="(Intercept)" &
coefficient!=0)

ggplot(coef_df_y1, aes(reorder(variable,
coefficient), coefficient)) +
  geom_col(fill="steelblue") + coord_flip()

y_y2 <- hypertension_clean$Y2
predictors_y2 <-
setdiff(names(hypertension_clean),
c("Y1","Y1_bc","Y2","X3"))
x_y2 <- model.matrix(~ .,
data=hypertension_clean[predictors_y2])[,-1]

set.seed(123)
lasso_y2 <- cv.glmnet(x_y2, y_y2, alpha=1,
family="binomial")

coef_df_y2 <- data.frame(
  variable = rownames(coef(lasso_y2,
s="lambda.min")),
  coefficient = as.numeric(coef(lasso_y2,
s="lambda.min"))
) %>% filter(variable!="(Intercept)" &
coefficient!=0)

ggplot(coef_df_y2, aes(reorder(variable,
coefficient), coefficient)) +
  geom_col(fill="tomato") + coord_flip()

numeric_only <- hypertension_clean %>%
select(where(is.numeric)) %>% select(-Y1)
cor_mat <- cor(numeric_only,
use="pairwise.complete.obs")

ggcorrplot(cor_mat, type="lower", lab=FALSE,
           colors=c("blue","white","red"))
```

Article

Appendix A

```
numeric_only <- hypertension_clean %>%
select(where(is.numeric)) %>%% select(-Y1)
cor_mat <- cor(numeric_only,
use="pairwise.complete.obs")

ggcorrplot(cor_mat, type="lower", lab=FALSE,
           colors=c("blue","white","red"))

df_num <- hypertension_clean %>%
select(where(is.numeric))

cor_y1 <-
sapply(df_num[setdiff(names(df_num),c("Y1","Y1_bc
","Y2")),
                  function(x) cor(df_num$Y1_bc, x))
top10_y1 <- names(sort(abs(cor_y1),
decreasing=TRUE))[1:10]

ggplot(data.frame(v=top10_y1, c=cor_y1[top10_y1]),
       aes(reorder(v,c), c, fill=c)) + geom_col()
+ coord_flip()

cor_y2 <-
sapply(df_num[setdiff(names(df_num),c("Y1","Y1_bc
","Y2")),
                  function(x) cor(df_num$Y2, x))
top10_y2 <- names(sort(abs(cor_y2),
decreasing=TRUE))[1:10]

ggplot(data.frame(v=top10_y2, c=cor_y2[top10_y2]),
       aes(reorder(v,c), c, fill=c)) + geom_col()
+ coord_flip()


plot_one <- function(v){
  x <- hypertension_clean[[v]]
  if(is.numeric(x) && length(unique(x))>10)
    ggplot(hypertension_clean,aes_string(x=v)) +
geom_histogram(bins=25)
  else
    ggplot(hypertension_clean,aes_string(x=v)) +
geom_bar()
}

plot_list <- lapply(names(hypertension_clean),
plot_one)

for(i in seq(1,length(plot_list),4)){

grid.arrange(grobs=plot_list[i:min(i+3,length(plo
t_list))], ncol=2, nrow=2)
}

model1 <- lm(Y1 ~ X1 + I(X1^2) + ns(X4,3) + X5 +
Xo + Xp + Xi + Xd,
             data=hypertension_clean)
model2 <- lm(Y1 ~ X1 + I(X1^2) + ns(X4,3) +
                X5 + Xo + Xp + Xi + Xd +
                Xr + Xq14 + Xq5 + Xq15,
             data=hypertension_clean)
model3 <- glm(
  Y2 ~ X1 + I(X1^2) + ns(X4,3) +
    X5 + Xo + Xi + Xp + Xd + Xq16 + Xq5 + Xq15,
  data=hypertension_clean,
  family=binomial)

ggplot(model1, aes(.fitted, .resid)) +
stat_binhex() + geom_hline(yintercept=0)
ggplot(model1, aes(sample=.stdresid)) + stat_qq()
+ stat_qq_line()

ggplot(model2, aes(.fitted, .resid)) +
stat_binhex() + geom_hline(yintercept=0)
ggplot(model2, aes(sample=.stdresid)) + stat_qq()
+ stat_qq_line()

prob3 <- predict(model3, type="response")
roc3 <- roc(hypertension_clean$Y2, prob3)
plot(roc3, col="blue")
auc(roc3)

newdata <- data.frame(
  X1 = seq(20,80,length=200),
  X4 = median(hypertension_clean$X4),
  X5 = median(hypertension_clean$X5),
  Xo=0, Xi=0, Xp=0, Xd=0,
  Xq16 = median(hypertension_clean$Xq16),
  Xq5 = median(hypertension_clean$Xq5),
  Xq15 = median(hypertension_clean$Xq15)
)


new_ns <-
as.data.frame(predict(ns(hypertension_clean$X4,3),
newdata$X4))
nd <- cbind(newdata, new_ns)

pred <- predict(model3, nd, type="link",
se.fit=TRUE)
invlogit <- function(z) exp(z)/(1+exp(z))

nd$prob <- invlogit(pred$fit)
nd$lower <- invlogit(pred$fit - 1.96*pred$se.fit)
nd$upper <- invlogit(pred$fit + 1.96*pred$se.fit)

ggplot(nd, aes(X1, prob)) +
  geom_line(color="blue") +
  geom_ribbon(aes(ymin=lower,ymax=upper),
alpha=0.3)

new_lm <- data.frame(
  X1 = seq(20,80,length=200),
  X4 = median(hypertension_clean$X4),
  X5 = median(hypertension_clean$X5),
  Xo=0, Xp=0, Xi=0, Xd=0
)


new_lm_ns <- cbind(new_lm,

as.data.frame(predict(ns(hypertension_clean$X4,3),
new_lm$X4)))

pred_lm <- predict(model1, new_lm_ns,
interval="confidence")

new_lm$fit <- pred_lm[,1]
new_lm$lower <- pred_lm[,2]
new_lm$upper <- pred_lm[,3]

ggplot(new_lm, aes(X1, fit)) +
  geom_line(color="red") +
  geom_ribbon(aes(ymin=lower,ymax=upper),
fill="pink", alpha=0.3)
```

Article

Appendix B

Note: According to the statement of Data availability, we are presenting the data situation of the first 10 samples in nearly 50 factors here.

X1	X2	X4	X5	Y1	Y2	Xa	Xb	Xc	Xd	Xe
71	146.5	26.14	0.81	140	1	1	1	2	1	1
62	154.0	25.59	0.82	140	1	1	1	3	0	0
59	163.1	19.26	0.83	120	0	1	1	5	0	0
73	147.0	27.63	0.87	141	1	1	1	2	0	0
76	147.5	25.05	0.96	123	0	1	1	2	0	0
60	152.8	28.95	0.83	153	1	1	1	1	0	0
60	144.5	28.93	0.88	110	0	1	1	3	0	0
85	144.3	25.86	0.97	127	0	1	1	1	0	0
67	145.0	21.50	0.86	144	1	1	1	1	0	0
69	144.5	18.06	0.87	108	0	1	1	3	0	0

Xf	Xg	Xh	Xi	Xj	Xk	Xl	Xm	Xn	Xo	Xp
0	0	0	0	0	0	1	1	0	1	0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	1	0	1	1
0	0	0	0	0	0	0	1	0	1	0
0	0	0	0	0	0	1	1	0	1	1
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	1	0	0	0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	1	0	0	0

Xq1	Xq2	Xq3	Xq4	Xq5	Xq6	Xq7	Xq8	Xq9	Xq10	Xq11	Xq12	Xq13	Xq14	Xq15	Xq16	Xq17	Xq18	Xq19	Xq20
4	4	3	2	2	2	2	4	1	2	2	2	3	3	1	3	1	1	1	1
4	3	2	2	2	2	2	4	1	1	2	4	2	1	3	4	1	1	1	1
4	5	5	2	4	4	3	2	1	1	2	5	5	2	4	4	1	1	1	1
5	5	3	3	2	3	2	3	2	2	3	3	2	5	2	3	2	2	1	1
5	1	2	4	2	2	2	5	2	1	2	5	2	3	1	5	1	3	1	5
4	4	4	3	2	2	2	5	1	2	2	5	3	1	1	3	1	1	1	1
4	3	3	2	3	3	3	4	1	1	1	5	3	4	1	2	1	1	1	1
5	4	3	3	2	2	2	3	1	1	1	5	2	1	1	4	1	1	1	1
4	4	3	3	3	2	3	2	2	2	2	4	3	2	3	4	2	2	1	1
4	2	2	4	3	2	2	5	1	1	1	5	1	1	5	3	2	1	1	1

Xr	Xs	Xt	Xu	Xv	Xw	Xx	Xy
1	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	1	2	1	0	0	0	0
0	1	3	1	0	0	0	0
0	1	2	1	0	0	0	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0
0	1	1	1	0	0	0	0